

SolSight

Solder Defect Detection System

Unsupervised Anomaly Detection for PCB Assembly

- **Fully convolutional autoencoder** for optical inspection
- Resolution-flexible anomaly detection
- No labeled real-world defects required for initial training

The Problem

PCB inspection is still constrained by the data problem.

- Manual solder-joint inspection is **slow and difficult to scale**.
- Defects are relatively rare compared with normal production.
- Supervised inspection systems require **representative labeled defect examples**.
- New boards, cameras, lenses, and manufacturing conditions introduce domain variation.

What if the system only needed to learn what a good solder joint looks like?

The Core Result

0.879

AUROC on real physical solder-joint data

Trained entirely on synthetic renders.

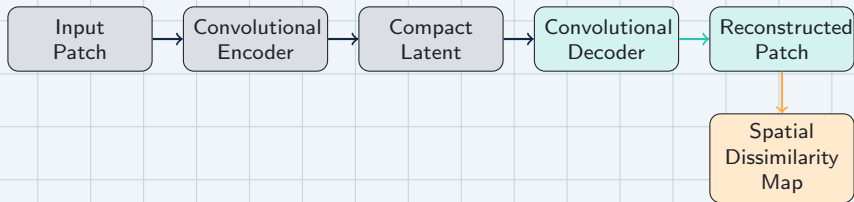
Tested **zero-shot** on real hardware it had never seen.

No labeled real-world defects were used for training.

The rest of this presentation answers three questions: **Why does it work?** **Does it generalize?**
Where does it fail?

Insight: Learn Normal, Flag Reconstruction Failure

A normal solder joint should reconstruct itself. An abnormal one should leave a measurable reconstruction error.



- Training uses **normal solder-joint examples** rather than an exhaustive defect taxonomy.
- The model is fully convolutional: **no dense layers tie it to one input resolution.**
- A custom differentiable SSIM implementation produces a **spatial anomaly map**, rather than only one global score.

Generalization Beyond Training Resolution

0.862

AUROC at 32×32 px

32 px was never used as a training tier. The model was trained on 16 px, 64 px and 128 px patches, yet achieved 0.862 AUROC at the unseen 32 px resolution.

Resolution	Training Exposure	AUROC
16 px	Trained	0.940
32 px	Zero-shot	0.862
64 px	Trained	0.963
128 px	Trained	0.850

Two independent forms of generalization now emerge: **unseen resolution** and **unseen physical hardware**.

Where It Breaks — And Why We Know

We did not only measure where the system works. We deliberately mapped where it stops working.

Dimension	Observed Safe Range	Breaking Region	Observed Cause
Spatial Resolution	16–160 px	≤ 14 px	SSIM window becomes too large
Defect Footprint	$\geq 1.5\%$ area	$< 0.5\%$	Global threshold becomes insensitive
Sensor Noise	$\sigma \leq 0.01$	$\sigma \geq 0.02$	SSIM contrast penalty

The important part is not that it fails. The important part is that the failure boundaries are **measurable and explainable**.

- High-resolution bridging recall exposes a **threshold calibration problem**, not an unexplained model collapse.
- Per-tier threshold calibration provides a practical mitigation.
- Stress testing establishes operating margins before deployment.

Real Hardware: Zero-Shot Domain Transfer

Synthetic training → physical solder joints

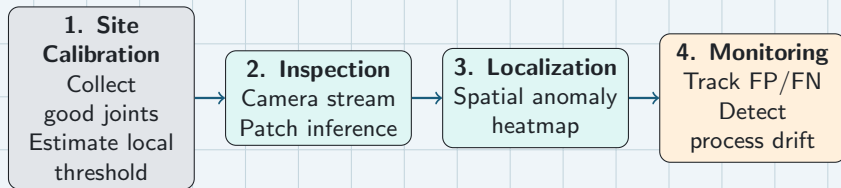
Metric	Result	Context
Real-World AUROC	0.879	Zero-shot synth → real
Normal-Joint FPR	0.0%	Site-calibrated threshold
Overall Defect Recall	48.5%	Site-calibrated threshold
Misaligned Component Recall	72.0%	Site-calibrated threshold
Excessive Solder Recall	46.0%	Site-calibrated threshold

Threshold calibration matters. The synthetic global threshold is below the real normal-joint baseline; applying it directly produces excess anomaly flags. Deployment uses $T_{\text{real}} = 0.2387$ calibrated on target hardware.

Validation dataset: SolDef_AI, Fontana, Calabrese et al., CC BY 4.0.

From Model to Factory Workflow

The deployment requirement is calibration, not defect labeling.



- Collect approximately 100–200 site-specific good samples for threshold calibration.
- Generate anomaly scores in real time (< 50 ms per patch) according to the current implementation.
- Expose the spatial DSSIM map so operators can inspect where the anomaly is occurring.

Where SolSight Fits

Capability	SolSight	Traditional AOI	Supervised DL
Defect labels for initial training	Not required	Typically required	Typically required
Resolution flexibility	Fully convolutional	System-dependent	Architecture-dependent
Synthetic → real transfer	0.879 AUROC	N/A as learning claim	Requires validation / adaptation
Fine-grained defect localization	Spatial heatmap	Inspection-dependent	Often strong
Tooling maturity	Emerging	Mature	Mature
Labeling burden	Low for initial training	Depends on system	Potentially high

The differentiation is specific: low initial labeling burden + resolution flexibility + zero-shot synthetic-to-real validation

Supervised approaches remain valuable where dense, fine-grained localization and mature production tooling are the primary requirements.

Technical Foundation

- **Language:** Python 3.10+
- **Deep Learning:** PyTorch 2.1+
- **Model:** Fully convolutional autoencoder
- **Similarity Metric:** Custom differentiable SSIM in PyTorch
- **Spatial Scoring:** Dynamic Gaussian windowing and spatial dissimilarity mapping
- **Evaluation:** ROC-AUC / AUROC, precision, recall and stress testing

Artifact	Current Value
Best checkpoint	1.17 MB
Maximum configured epochs	30
Configured batch size	64
Inference target	<50 ms / patch

Early stopping means the effective training duration can be shorter than the configured 30-epoch maximum.

What We Can Reproduce Today

The repository is part of the evidence.

Evidence	Current Repository State
Synthetic patches	2,100
Best model checkpoint	1.17 MB
Real-world AUROC	0.879
Zero-shot 32 px AUROC	0.862
Site-calibrated real FPR	0.0%

Reproducibility principle

The presentation reports the **current committed evaluation artifacts** rather than numbers from an earlier experiment.

Evaluation outputs include model metrics, threshold configuration and stress-test summaries.

What Comes Next

- **Edge Deployment**

- ◆ ONNX export and TVM-based optimization for ARM / FPGA targets.

- **Multi-Modal Inspection**

- ◆ Combine visible and thermal imaging for additional defect evidence.

- **Online Adaptation**

- ◆ Refine thresholds using controlled operator feedback and production statistics.

- **Explainability**

- ◆ Develop richer defect-localization and importance visualization.

The immediate objective is not a larger model. It is a more reliable operating envelope.

SolSight

Teaching inspection systems what “normal” looks like.

0.879 AUROC

Zero-shot transfer from **synthetic training** to **real solder-joint data**.

Questions?

Resources & Attribution

GitHub Repository

`github.com/stanmaluki/SolderDefectDetectionSystem`

Validation Dataset

SolDef_AI Fontana, Calabrese et al. CC BY 4.0

Software

MIT License

Dataset attribution and licensing are retained explicitly to distinguish project results from third-party validation data.